H&M Teams System

Software Requirements Specification (SRS)

Group 2, Case Study 2

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Change History

The table below highlights key changes made to this Project Plan document since its inception:

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Table of Contents

Introduction……6

Project Name……..6

Scope……..6

Objectives……...7

Definitions, Acronyms, and Abbreviations……7

References……..8

Overview……8

Overall Description……8

Product Perspective……8

System Interfaces……...8

User Interfaces……9

Hardware Interfaces……...9

Software Interfaces……9

Communications Interfaces……..10

Memory Constraints……10

Operations……10

Site Adaptation Requirements……10

Product Functions……10

User Characteristics……11

Constraints……12

Assumptions and Dependencies……...12

Specific Requirements……13

Functional Requirements……..13

UC1: Logging In……..13

UC2: Maintaining User Data……14

UC3: Uploading Assignments……..16

UC4: Uploading Course Material……18

UC5: Viewing Submitted Assignments……20

UC6: Submitting Assignments……22
UC7: Viewing Course Material……..23
Non-functional Requirements……25
Performance Requirements……25
Security Requirements……25
Reliability Requirements……26
Availability Requirements……..26
Usability Requirements……..27
Maintainability Requirements……27
System Functions……28
Sub-Functions for Authentication and Access Control……..28
Sub-Functions for User and Course Data Management……28
Sub-Functions for Course Material Management……..28
Sub-Functions for Assignment Management……..29
Sub-Functions for Assignment Submission and Version Control……..29
Sub-Functions for Grading and Feedback……29
Supplementary Material……30
Data Dictionary……..30
Use Case Diagram……..35
Domain Model……36

1. Introduction:

1.1 Project Name

"H&M Teams System"

1.2 Purpose

The main purpose of **H&M Teams System** is to improve the efficiency of managing academic activities at Quaid-i-Azam University by providing a single user-friendly platform. This system will perform the following functions:

- Provide a secure, centralized, and access-based platform for teachers and students to manage course-related activities, including lectures, assignments, and submissions.
- Contain the most essential and current information about students (full names, unique identification numbers, semester, courses, etc.), teachers (full names, courses they are teaching, etc.), and courses (IDs, names, students enrolled, etc.).
- Enable systematic access to course material by allowing students to conveniently view and retrieve course materials and assignments for their registered courses.
- Support organized assignment submission by allowing students to submit their work electronically within the allocated deadline and retain multiple versions of a submission.
- Improve accountability and transparency through computerized records of course registrations, uploaded materials, assignment submissions, and submission times.
- Automate deadline management and notifications to keep students informed about newly uploaded assignments and relevant submission deadlines.

1.3 Scope

- Provide varying levels of access to each user based on their role in the system.
- Give administrators the authority to add and maintain lists of courses, students, and teachers for each semester.
- Teachers easily upload assignments and manage course materials for their courses through the system.
- Students view course materials and have the option to submit multiple versions of their assignments.
- Students are automatically notified about uploaded assignments and upcoming deadlines.
- Teachers review and grade submitted assignments, providing short feedback if necessary.

- Students are automatically notified when a non-PDF file is attached to an assignment submission, and the system offers to convert it to PDF before final submission.
- Students can open a blank document inside the system, type their assignment there, and submit it directly as their assignment file.

1.4 Objectives

The goal is to develop an all-encompassing and user-friendly platform that improves student-teacher interaction within the university. The key objectives include:

- Developing a centralized and secure system for efficient communication between teachers and students which curbs issues related to scattered resources and missing assignments.
- Ensuring administrators maintain the most current database of courses, students, and teachers for each semester so that outdated information is removed.
- Facilitating students by providing convenient access to the course materials and assignments of their enrolled courses.
- Providing flexibility to students in the submission of assignments to improve the quality of work turned in.
- Automatically alerting students regarding submission deadlines to reduce the probability of missing assignments.
- Allow teachers to view and grade student work, as well as provide feedback on a single platform to reduce paper usage for conserving the environment.

1.5 Definitions, Acronyms, and Abbreviations

- **Access Control:** The system's mechanism to restrict access to authorized users based on their roles (student, supervisor, administrator).
- **Administrator:** A system manager responsible for entering and maintaining the data of courses, students, and teachers.
- **Authentication:** The process of verifying a user's identity via email verification before granting access to the system.
- **Course Materials:** This is an amalgamation of books, lectures, notes, slides, and other relevant study materials.
- **Gantt Chart:** A visual representation of the project schedule, showing tasks, durations, and dependencies.
- **Student:** A final-year student who registers a project, submits work products, and interacts with the supervisor through the system.
- **Teacher:** A faculty member responsible for uploading course materials and grading assignments.

1.6 References

- IEEE Standard for Software Requirements Specifications (IEEE Std 830-1998).
- ISO/IEC/IEEE 16326:2019 Guidelines for Life Cycle Processes – Project Management.
- Department of Computer Science Project Guidelines, [University/Department internal documentation].

1.7 Overview

This document is organized to provide a comprehensive understanding of the H&M Teams System. The SRS is divided into several main sections:

i. **Introduction:** Describes the purpose, scope, definitions, references, and an overview of the SRS.
ii. **Overall Description:** Provides details on the product perspective, functionality, user characteristics, constraints, and assumptions.
iii. **Specific Requirements:** Outlines in detail the functional requirements (use cases) and non-functional requirements, including system functions and performance criteria.
iv. **Supplementary Material:** Includes supporting documentation such as the data dictionary, domain model, and use case diagram.

2. Overall Description:

2.1 Product Perspective

H&M Teams System is an independent, self-contained, web-based system developed for the Department of Computer Sciences at Quaid-i-Azam University. It is not a component of any larger system and does not depend on any other university software to perform its functions. It is intended to replace the informal use of general-purpose messaging applications (such as WhatsApp) for managing university work, and to serve as a single centralized platform for communication and collaboration between students and teachers.

Unlike generic learning or messaging tools, H&M Teams System is tailored to the academic needs of the department. It maintains up-to-date records of courses, students, and teachers for each semester, enables teachers to upload course materials and assignments, and allows students to view and download course materials. Students can submit multiple versions of their assignments before the deadline, either by attaching a file or by typing their work in a blank document opened inside the system. If an attached file is convertible to a PDF, the system notifies the student and offers to convert it to PDF before final submission. The system also supports grading and feedback on a single platform. Access to all functionality is controlled by user role (Administrator, Teacher, or Student).

2.1.1 System Interfaces

- H&M Teams System operates as a standalone system and has no interfaces with other university systems.

- User authentication is handled within the system through user IDs and passwords.
- All data on courses, students, teachers, course materials, assignment submissions (including multiple versions), grades, and feedback is entered and stored within the system's own database.
- File format conversion (non-PDF to PDF) and the built-in document editor are provided by the system itself and do not rely on any external university system.

2.1.2 User Interfaces

The system shall provide a web-based graphical user interface accessible through commonly used web browsers on desktop and mobile devices.

The interface shall be simple to navigate and lightweight, so that it remains usable on slower Internet connections.

Each user shall be presented with only the functions permitted for their role:

- *Administrator:* adding and maintaining course, student, and teacher information for each semester.
- *Teacher:* uploading course materials and assignments, viewing and grading submissions, and giving feedback.
- *Student:* viewing and downloading course materials; submitting assignments (including multiple versions) by attaching a file or by typing in a blank document opened within the system; receiving notifications about new assignments, upcoming deadlines, and non-PDF attachments; and choosing whether to convert a non-PDF attachment to PDF before final submission.

2.1.3 Hardware Interfaces

- The system has no direct hardware interfaces.
- Users access the system through standard devices (computers, laptops, tablets, and smartphones) with a browser and an Internet connection.
- The system is hosted on a server that stores the database and uploaded files.

2.1.4 Software Interfaces

- Front end: HTML, CSS, and JavaScript, including the in-system document editor used by students to type their assignments.
- Database: MySQL, which stores course, student, and teacher data and all versions of submitted work.
- Back end: a dedicated server-side application that handles requests from the user interface, enforces role-based access control, performs file format checking and PDF conversion, and communicates with the database.
- Client software: a commonly used web browser that supports the system's front end.

2.1.5 Communications Interfaces

- Communication between the user's browser and the back-end server shall take place over the Internet using standard web protocols (HTTP/HTTPS).
- Data transmitted between users and the system shall be encrypted to protect user data.
- The system shall send notifications to students about newly uploaded assignments and relevant submission deadlines.
- The system shall automatically notify a student when a non-PDF file is attached to an assignment submission.

2.1.6 Memory Constraints

- To ensure efficient use of storage, each file submission shall not exceed 2 MB.
- Common assignment and document file formats, including source code file formats, shall be supported.
- An assignment typed within the system shall be submitted as the student's assignment file and is subject to the same 2 MB limit.
- A file generated by converting a non-PDF attachment to PDF shall also remain within the 2 MB limit.

2.1.7 Operations

- The system shall be available to users throughout the academic semester.
- The administrator shall update the lists of courses, students, and teachers at the start of each semester, so that outdated information is removed.
- Teachers shall set assignment deadlines, which can be changed once.
- Only students registered in a course shall be able to access its materials and assignments.
- Students may submit multiple versions of an assignment before the deadline. A version may be an attached file or a document typed within the system.
- When a student attaches a non-PDF file, the system shall offer to convert it to PDF before final submission. The student may accept the conversion or proceed with the original file.
- Regular backups of the database and uploaded files shall be maintained.

2.1.8 Site Adaptation Requirements

- The system shall comply with university policies and regulations on data security and privacy.
- Course, student, and teacher data for each semester shall be loaded by the administrator before the system is used.

2.2 Product Functions

H&M Teams System includes the following key functions:

- **User Authentication and Access Control:** Verifies every user's identity through email verification and multi-factor authentication before granting access. Gives each user access only to the functions permitted for their role (Administrator, Teacher, or Student). Ensures that only authorized users can access course data and materials.
- **Maintenance of Student, Teacher, and Course Information:** The administrator adds and maintains the lists of students, teachers, and courses for each semester. Includes the students enrolled in each course and the teacher assigned to it. Maintains a central, up-to-date record and allows outdated information to be removed at the start of each semester.
- **Uploading and Managing Course Materials:** Teachers upload and manage course materials (such as lectures, notes, and slides) for the courses they teach. Materials are available only to students registered in the respective course.
- **Viewing and Downloading Course Materials:** Students view the course materials of their registered courses. Students may also download the materials for their own use. Replaces the scattered sharing of materials through messaging applications with a single organized location.
- **Assignment Upload and Deadline Management:** Teachers upload assignments for their courses. Teachers specify a submission deadline for each assignment, which can be changed once. The system automatically notifies students about newly uploaded assignments and upcoming deadlines.
- **Assignment Submission with Multiple Versions:** Students submit assignments electronically before the deadline. Multiple versions of the same assignment can be uploaded. The system records the submission time of each version, so teachers can review progress over time. Students can submit in either of two ways: Attaching a file: in a supported format, up to 2 MB or typing within the system: the student opens a blank document inside the system, types the assignment there, and submits it directly as the assignment file.
- **PDF Conversion of Attached Files:** The system automatically notifies the student when a non-PDF file is attached to an assignment submission. The system offers to convert the file to PDF before final submission.
- **Viewing, Grading, and Feedback on Submissions:** Teachers view the assignments submitted by students for their courses, including all submitted versions. Teachers grade the submitted assignments. Teachers may provide short feedback where necessary to help students improve their work.
- **Records and Accountability:** Keeps computerized records of course registrations, uploaded materials, assignment submissions, and submission times. Improves accountability and transparency for both students and teachers.

2.3 User Characteristics

- **Administrator:** Department staff with at least a bachelor's degree and moderate computer skills, who enter and maintain student,

teacher, and course data mainly at the start of each semester. As they use the system infrequently and need not be technical experts, the interface must be simple and easy to relearn.

- **Teacher:** Faculty members with a master's or doctoral degree and moderate to high computer skills, who currently rely on messaging apps to share materials. They have limited time, so uploading, grading, and reviewing submissions must be quick and require little training.
- **Student:** Undergraduate students with generally high technical skills who often work under deadline pressure and revise their work several times. Since they use different software, devices, and Internet speeds, the system must support multiple file formats and versions, send reminders, and stay lightweight.

2.4 Constraints

- The system must be designed and developed within the allocated project timeline of the current academic semester (Fall 2026).
- The system must comply with university policies and regulations pertaining to data security and privacy.
- The system must provide simple navigation, be lightweight and compatible with commonly used web browsers and devices.
- The system must support common assignment and document file formats, including source code file formats, with a maximum file size of 2 MB per submission for efficient memory usage.
- The system must implement multi-factor authentication and encryption features to protect user data.
- The platform must be lightweight in nature to allow its usage with slower internet connections.

2.5 Assumptions and Dependencies

- Every student and teacher using the system is registered, has provided correct information when creating accounts, and has valid login credentials.
- Students and teachers are assumed to have a stable Internet connection as well as Browsers which support the system to access the website.
- The system assumes that the list of courses and the students registered in each course is provided and maintained accurately by the administrator.
- Teachers are assumed to upload lectures and assignments for the courses they teach.
- Only students registered for a particular course are assumed to have access to its course materials.
- Teachers are responsible for specifying the allocated submission deadline for each assignment which can be changed once.

3. Specific Requirements

3.1 Functional Requirements (Use Cases)

3.1.1 UC1: Logging In

- **Actor:** Student, Teacher, Administrator
- **Inputs:**
  - User: userID, password.
- **Outputs:**
  - Active Session.
- **Pre-conditions:**
  1. The user has an active account in the system, created by the administrator.
  2. The user is not already logged into the system.
- **Main Success Scenario:**
  1. User requests to log into the system.
  2. System asks the user to provide their login credentials.
  3. User enters their user ID and password.
  4. System verifies the credentials.
  5. System identifies the role of the user (student, teacher, or administrator).
  6. System starts a session for the user and displays the home page with the features available to that role.
- **Alternative Flows:**
  - Alt Flow 3a:
    User has forgotten their password and requests to reset it. The system verifies the user through their registered email and allows the user to set a new password.
  - Alt Flow 4a:
    User is logging in for the first time with a temporary password. The system asks the user to set a new password before proceeding.
- **Exception Flows:**
  - Exception 1:
    User enters an incorrect user ID or password. The system displays an "Error: Invalid Credentials" message and allows the user to try again.
  - Exception 2:
    User leaves the user ID or password empty. The system displays an "Error: Please enter your user ID and password" message.
  - Exception 3:

User's account has been marked as inactive, for example because the student or teacher has left. The system displays an error message and does not allow login.

- Exception 4:
  User enters incorrect credentials multiple times in a row. The system temporarily blocks further attempts and displays an error message.

- **Post-conditions:**
  1. The user is logged into the system and a session is started.
  2. The user has access only to the features allowed for their role.

3.1.2 UC2: Maintaining User Data

- **Actor:** Administrator
- **Inputs:**
  - Administrator: adminName, adminEmail.
  - Student: studentName, studentEmail, studentDOB, studentContact, semester, program, department.
  - Teacher: teacherName, teacherEmail, teacherContact, rank, department.
  - Course: courseID, courseName, courseDescription, semester.
- **Outputs:**
  - Updated record of teachers and students for a semester.
  - Course Lists: students registered for the course, teacher assigned to the course.
  - Update Confirmation: data saved successfully.
- **Pre-conditions:**
  1. Administrator is logged into the system.
  2. The semester for which the data is being maintained exists in the system.
- **Main Success Scenario:**
  1. Administrator requests to maintain user data.
  2. System displays the categories of data that can be maintained, namely students, teachers, and courses.
  3. Administrator selects the desired category.
  4. System displays the list of existing records of the selected category for the current semester.
  5. Administrator requests to add a new record.
  6. System asks the administrator to provide the details of the record.
  7. Administrator enters the required details.
  8. System verifies that the details are complete and that the record does not already exist.

9. System saves the record and displays a confirmation message along with the details of the new record.

- **Alternative Flows:**
  1. Alt Flow 4a:
     Administrator searches for a record by name or ID. The system displays the matching records.
  2. Alt Flow 5a:
     Administrator selects an existing record and edits its details. The system saves the changes and displays the updated record.
  3. Alt Flow 5b:
     Administrator marks a student or teacher as having left the institution. The system changes the status of the record to inactive, removes the user from the current course lists, and retains the record and its previous data.
  4. Alt Flow 5c:
     Administrator removes a course from the semester. The system removes the course from the current semester and retains its previous data.
  5. Alt Flow 5d:
     Administrator registers a student for a course. The system adds the student to the list of students registered for that course.
  6. Alt Flow 5e:
     Administrator removes a student's registration from a course. The system removes the student from the list of students registered for that course.
  7. Alt Flow 5f:
     Administrator assigns a teacher to a course. The system saves the assignment and adds the course to the teacher's list of courses.
  8. Alt Flow 5g:
     Administrator changes the teacher assigned to a course. The system replaces the previous assignment with the new one.
  9. Alt Flow 7a:
     Administrator decides to complete the update later and exits the system. The system retains the entered details and displays them when the administrator re-enters the system, allowing the update to be completed.
- **Exception Flows:**
  - Exception 1:
    Administrator tries to add a record that already exists. The system displays an error message and does not add the record.
  - Exception 2:

Administrator leaves a required detail empty or enters an invalid detail. The system displays an error message and asks the administrator to correct the details.

- Exception 3:
  Administrator tries to register a student for a course that the student is already registered for. The system displays an error message.
- Exception 4:
  Administrator tries to assign a teacher to a course that already has an assigned teacher and does not choose to replace them. The system displays an error message.

- **Post-conditions:**
  1. The data of students, teachers, or courses is added, updated, or marked inactive as requested.
  2. Course registrations and teacher assignments are updated.
  3. Users marked as inactive can no longer access the system, but their previous data is retained.
  4. The updated data is available to the users according to their roles.

3.1.3 UC3: Uploading Assignments

- **Actor:** Teacher
- **Inputs:**
  - Teacher: teacherName, teacherEmail, teacherContact, rank, department.
  - Course: courseID, courseName, courseDescription.
- **Outputs:**
  - Assignment: assignmentID, assignmentTitle, assignmentInstructions, assignmentDeadline, marks, referenceMaterial, notificationMessage.
  - Recipients: recipientID, recipientName, recipientEmail.
- **Pre-conditions:**
  1. Teacher is logged into the system.
  2. Teacher is assigned to the course that they want to upload the assignment of.
- **Main Success Scenario:**
  1. Teacher sees a list of all the courses they are teaching and selects the course that they want to upload the assignment of.
  2. Teacher navigates to the assignments tab.
  3. Teacher clicks on the create assignment option.
  4. System opens the assignment creation page.
  5. Teacher enters the title of the assignment in the title field.

6. Teacher enters the necessary instructions for the assignment in the instructions field.
7. Teacher clicks on the option to attach any reference materials for the assignment.
8. System displays a list of options for the teacher to choose from, such as uploading from device, adding links, or opening a blank document.
9. Teacher clicks on the uploading from device button.
10. System opens the folders in the teacher's device.
11. Teacher chooses the desired files for uploading.
12. Teacher enters the assignment marks in the marks field.
13. Teacher sees options to assign to all students or specific students.
14. Teacher clicks on the assign to all students option.
15. Teacher enters the due date and time of the assignment in the deadline fields.
16. Teacher clicks on the assign button.

- **Post-conditions:**
  1. The assignment is saved in the system under the selected course, along with its title, instructions, marks, deadline, and reference material.
  2. The assignment is available to the students it was assigned to (all registered students or only the selected ones).
  3. The assigned students are notified about the new assignment and its deadline.
- **Alternative Flows:**
  - Alt Flow 7a:
    Teacher does not add any reference material. System allows the teacher to proceed anyway.
  - Alt Flow 8a:
    Teacher clicks on the option to add links instead and the system displays a text box where links can be pasted as reference materials.
  - Alt Flow 8b:
    Teacher clicks on the option to open a blank document and the system displays a blank document with simple text editing options where the teacher can type assignment instructions.
  - Alt Flow 13a:
    Teacher clicks on assign to selected students option. The system displays a list of all students enrolled for the course and allows the teacher to select one or more students for the assignment.
  - Alt Flow 16a:
    Teacher decides to upload the assignment later and exits the system. The system retains the assignment details and displays

them when the teacher re-enters the system, allowing uploading.

- **Exception Flows:**
  - Exception 1: Teacher sets a deadline earlier than the current time and clicks on the “Assign” button. The system displays an error message.

3.1.4 UC4: Uploading Course Material

- **Actor:** Teacher
- **Inputs:**
  - Teacher: teacherName, teacherEmail, teacherContact, rank, department.
  - Course: courseID, courseName, courseDescription.
  - Course Material: materialTitle, materialDescription, materialFiles.
- **Outputs:**
  - Course Material: cmID, cmTitle, materialDescription, uploadDate, uploadTime.
  - Upload Confirmation: course material uploaded successfully.
- **Pre-conditions:**
  1. Teacher is logged into the system.
  2. Teacher is assigned to the course that they want to upload the course material of.
- **Main Success Scenario:**
  1. Teacher sees a list of all the courses they are teaching and selects the course that they want to upload the course material of.
  2. System displays the options available for the selected course, including course material.
  3. Teacher clicks on the option for course material.
  4. System displays the list of course materials already uploaded for the course, along with the option to upload new course material.
  5. Teacher requests to upload new course material.
  6. System asks the teacher to provide the details of the course material.
  7. Teacher enters the title and description of the course material.
  8. Teacher chooses to attach files and selects the desired files from their device.
  9. System displays the list of attached files.
  10. Teacher confirms the upload.

- **Post-conditions:**
  1. The course material is saved in the system under the selected course.
  2. The course material is available for viewing by the students registered for the course.
  3. Students registered for the course are notified about the newly uploaded course material.
  4. The upload date and time of the course material are recorded.
- **Alternative Flows:**
  - Alt Flow 7a:
    Teacher does not enter a description. The system allows the teacher to proceed with only the title.
  - Alt Flow 8a:
    Teacher chooses to add a link. The system allows the link to be entered as course material.
  - Alt Flow 8b:
    Teacher chooses to create the course material in the system itself. The system provides a blank document with simple text editing options where the teacher can type the content.
  - Alt Flow 9a:
    Teacher attaches more files. The system displays the complete list of attached files.
  - Alt Flow 9b:
    Teacher removes a file from the attached files. The system removes that file and retains any other attached files.
  - Alt Flow 10a:
    Teacher decides to upload the course material later and exits the system. The system retains the course material details and attached files, and displays them when the teacher re-enters the system, allowing uploading.
- **Exception Flows:**
  - Exception 1:
    Teacher confirms the upload without entering a title. The system displays an error message.
  - Exception 2:
    Teacher confirms the upload without attaching any file or link. The system displays an error message.
  - Exception 3:
    Teacher attaches a file that is of an unsupported type or exceeds the allowed size. The system displays an error message and does not attach the file.
  - Exception 4:

The upload fails due to a system or connection problem. The system displays an error message, retains the entered details, and allows the teacher to try again.

3.1.5 UC 5: Viewing Submitted Assignments

- **Actor:** Teacher
- **Inputs:**
  - Teacher: teacherName, teacherEmail, teacherContact, rank, department.
  - Course: courseID, courseName, courseDescription.
  - Assignment: assignmentID, assignmentTitle, assignmentInstructions, assignmentDeadline, marks.
- **Outputs:**
  - Submission: submissionID, studentID, studentName, submissionDate, submissionTime, submissionVersion, submittedFiles.
  - Grade: obtainedMarks, feedback, gradeDate, gradeTime.
- **Pre-conditions:**
  1. Teacher is logged into the system.
  2. Teacher is assigned to the course whose submitted assignments they want to view.
  3. The assignment has been uploaded for the course.
  4. At least one student has submitted the assignment.
- **Main Success Scenario:**
  1. Teacher sees a list of all the courses they are teaching and selects the desired course.
  2. System displays the options available for the selected course, including the assignments tab.
  3. Teacher clicks on the assignments option.
  4. System displays the list of assignments uploaded for the course.
  5. Teacher selects the desired assignment.
  6. System displays the list of students who have submitted the assignment, along with the submission date and time of each student.
  7. Teacher selects the submission of a particular student.
  8. System displays the latest version of the submission by default, along with the total number of versions submitted.
  9. Teacher opens the submitted files.
  10. System displays the content of the submitted files to the teacher.
  11. Teacher enters the marks the student has obtained for the submission.

12. System saves the marks against the submission and displays them student.

- **Post-conditions:**
  1. The submitted files and all versions of the submission remain unchanged in the system.
  2. The marks entered by the teacher are saved against the student's submission and are available to the student.
  3. If feedback is given, it is saved and available to the student.
- **Alternative Flows:**
  - Alt Flow 6a:
    The teacher searches for a student by name or ID. The system displays the matching submissions.
  - Alt Flow 8a:
    The teacher selects an earlier version of the submission. The system displays the files of the selected version, along with its submission date and time.
  - Alt Flow 8b:
    The student has submitted only one version. The system displays that version only.
  - Alt Flow 10a:
    Teacher requests to download the submitted files. The system saves a copy of the files on the teacher's device.
  - Alt Flow 12a:
    Teacher chooses to give feedback to the student. The system allows the teacher to enter a short feedback and saves it.
  - Alt Flow 12b:
    Teacher decides to grade the assignment later and exits the system. The system retains the submissions and displays them when the teacher re-enters the system, allowing grading.
- **Exception Flows:**
  - Exception 1:
    The selected course has no uploaded assignments. The system displays an error message.
  - Exception 2:
    No student has submitted the selected assignment. The system displays an error message.
  - Exception 3:
    The selected file cannot be opened or loaded. The system displays an error message and allows the teacher to try again.

3.1.6 UC6: Submitting Assignments

- **Actor:** Student

- **Inputs:**
  - Student: studentName, studentEmail, studentDOB, studentContact, semester, program, department.
  - Course: courseID, courseName, courseDescription.
  - Assignment: assignmentID, assignmentTitle, assignmentInstructions, assignmentDeadline, recipients, marks, referenceMaterial.
- **Outputs:**
  - Submission: submissionID, submissionDate, submissionTime, submittedFiles, submissionVersion, submissionConfirmation.
- **Pre-conditions:**
  1. Student is logged into the system.
  2. Student is registered for the course that they want to submit the assignment of.
- **Main Success Scenario:**
  1. Student sees and clicks the attach files option.
  2. System displays a list of options for the student to choose from for uploading.
  3. Student clicks on the upload from device option.
  4. System opens the folders in the student's device.
  5. Student chooses the desired files for submission.
  6. System displays a loading bar while it loads the file selected by the student.
  7. System displays options to submit, review, remove, and convert the submission to a PDF file.
  8. Student clicks on the submit option.
- **Post-conditions:**
  1. The submitted files are saved in the system against the assignment, with the submission date and time recorded.
  2. The submission is stored as a new version, and all previous versions of the same assignment are retained.
  3. The submitted assignment is available for the teacher to view and grade.
  4. The system displays a success message with the submission's details, date, and time.
- **Alternative Flows:**
  - Alt Flow 1a:
    Student clicks on the file attachment option again and adds more files to be submitted. The system displays the complete list of attached files.

- Alt Flow 2a:
  Student clicks on taking photo option and the system opens the device camera. The student has options to retake and proceed for taking photos.
- Alt Flow 2b:
  Student clicks on adding links option and the system displays a text box where links can be pasted for submission.
- Alt Flow 2c:
  Student clicks on opening a blank document option and the system displays a blank document with simple text editing options where the student can type the assignment.
- Alt Flow 7a:
  Student clicks on the conversion to PDF option and the system converts the file to a PDF if supported.
- Alt Flow 8a:
  Student decides to submit the assignment later and exits the system. The system retains the attached files and displays them when the student re-enters the system, allowing submission.
- Alt Flow 8b:
  Student clicks on the submitting another version option and attaches a newer submission of the assignment before the deadline. The system saves the version number and the attached files against it along with the submission date and time.
- Alt Flow 8c:
  Student clicks on the viewing submission history option and views all their submissions for the assignment along with their attached files and submission times.

- **Exception Flows:**
  - Exception 1:
    The student tries uploading the assignment after the deadline and the system displays an error message.
  - Exception 2:
    The student selects a file larger than 2 MB. The system rejects it and displays a file-size error, and the student may choose another file.
  - Exception 3:
    The submission deadline passes before the student submits their work. The system does not accept the submission and displays a deadline-passed message.

3.1.7 UC 7: Viewing Course Material

- **Actor:** Student
- **Inputs:**

  - Student: studentName, semester, program, department.
  - Course: courseID, courseName, courseDescription.
  - Course Material: cmID, cmTitle, uploadDate, uploadTime.
- **Outputs:**
  - Course Material: materialTitle, materialDescription, uploadDate, materialFiles.
  - Viewing Confirmation: material displayed to the student.
- **Pre-conditions:**
  1. Student is logged into the system.
  2. Student is registered for the course whose material they want to view.
  3. Teacher has uploaded at least one course material for the course.
- **Main Success Scenario:**
  1. Student requests to view the list of courses they are registered for.
  2. System displays the list of the student's registered courses for the current semester.
  3. Student selects the desired course.
  4. System displays the options available for the selected course, including an option for course material.
  5. Student selects the course material option.
  6. System displays the list of course materials uploaded for that course, along with their titles and upload dates.
  7. Student selects the desired course material.
  8. System displays the details of the selected course material and the material itself.
  9. Student opens the attached file.
- **Post-conditions:**
  1. The selected course material is displayed to the student.
  2. The course material remains unchanged in the system.
- **Alternative Flows:**
  - Alt Flow 6a:
    Student searches for a course material by its title. The system displays the course materials matching the search.
  - Alt Flow 9a:
    Student requests to download the file. The system saves a copy of the file on the student's device.
  - Alt Flow 10a:
    Student exits the system while viewing the course material. The system retains the student's session details and allows the student to view the course material again on re-entering.

- **Exception Flows:**
  - Exception 1:
    The student has no registered courses for the semester. The system displays an error message.
  - Exception 2:
    The selected course has no uploaded course material. The system displays an error message.
  - Exception 3:
    The selected file cannot be opened or loaded. The system displays an error message and allows the student to try again.

3.2 Non-functional Requirements

3.2.1 Performance Requirements

- The system shall support access from any standard device (computer, laptop, tablet, smartphone) with a web browser, with no fixed limit on the number of terminals.
- The system shall support at least 200 simultaneous users without performance degradation.
- The system shall handle the records of at least 1,500 registered users and the courses, materials, and assignment submissions (including all versions) of an academic semester.
- Each submitted file shall not exceed 2 MB.
- Login and authentication shall complete within 3 seconds.
- Pages (course materials, assignments, submissions) shall load within 3 seconds under normal Internet conditions.
- Uploading a file of up to 2 MB, including saving a new version, shall complete within 5 seconds.
- Converting a non-PDF file to PDF shall complete within 10 seconds.
- Retrieving the list of previous versions shall complete within 3 seconds.
- Saving a grade and feedback shall complete within 2 seconds.
- Notifications about new assignments, deadlines, and non-PDF attachments shall be delivered within 1 minute of the triggering event, except the non-PDF notification, which shall appear immediately on attachment.
- The system shall remain usable on slower Internet connections.

3.2.2 Security Requirements

- All data transmitted between users and the system shall be encrypted using HTTPS.
- All stored user data, including submissions and grades, shall be encrypted, and passwords shall be stored only in hashed form.

- The system shall require email verification and multi-factor authentication at login.
- The system shall enforce role-based access control.
- The system shall record the time of each submission using server time, so that students cannot alter it.
- Once submitted, a version shall not be modifiable or deletable by the student, and previous versions shall be retained unchanged.
- The system shall keep logs of logins, uploads, submissions, grade changes, and administrator changes to user data.
- The system shall check the type and size of every uploaded file before accepting it, and shall verify the integrity of stored files and critical data (such as grades and deadlines).
- Sessions shall expire after a period of inactivity.

3.2.3 Reliability Requirements

- A submission shall be saved completely or not at all, so that a failed upload or system error never creates a partial or corrupted version.
- The system shall never lose or overwrite previously submitted versions when a new version is submitted.
- If PDF conversion fails, the student's original file shall remain unaffected and available for submission.
- Files attached but not yet submitted, and documents typed within the system, shall be retained if the student exits or the connection is lost.
- The system shall enforce deadlines consistently using server time, including any single deadline change by the teacher.
- Notifications shall be generated for every new assignment and approaching deadline.
- The system shall validate user input (such as marks and file types) and display clear error messages when an operation fails.

3.2.4 Availability Requirements

- The system shall be available at least 99.5% of the time during the academic semester.
- Maintenance shall be scheduled outside peak periods and shall not take place close to assignment deadlines.
- Automatic backups of the database and uploaded files shall be performed at least daily and stored in a separate secure location.
- Typed documents shall be saved automatically at regular intervals (checkpoints), so that at most a few minutes of work is lost after a failure.

- After a system failure, the system shall be restored from the latest backup and restarted within 4 hours, with no loss of data submitted before the last backup.

3.2.5 Usability Requirements

- The interface shall be simple and intuitive, with clear navigation for each role.
- Forms shall include inline validation so users can correct errors immediately.
- The system shall provide tooltips and help for key functions, such as attaching files, converting to PDF, and submitting new versions.
- The interface shall be responsive and usable on desktop and web browsers on mobile devices.

3.2.6 Maintainability Requirements

- The system shall be built with an object-oriented design, using self-contained classes so that a change to one part has minimal impact on the rest of the system.
- The system shall be divided into separate modules (for example, user management, course materials, assignment submission, grading, and notifications) with clearly defined interfaces between them.
- The front end, back end, and database shall be kept separate, so that any one can be modified without rewriting the others.
- The source code shall follow consistent naming conventions and coding standards, and shall include comments explaining the purpose of each class and method.
- All source code shall be managed in a version control system (GitHub), with a record of every change.
- The design shall remain traceable to the requirements. Each class and module shall be traceable to the design and to the SRS, so the effect of a requirement change can be identified quickly.
- Settings that are likely to change, such as the maximum file size (2 MB), the supported file formats, session timeout, and notification timing, shall be configurable without changing the program code.
- Adding a new supported file format or changing the conversion method shall not require changes to unrelated modules.
- The system shall maintain error and activity logs so that faults can be located and corrected quickly.

- Each module shall be tested individually, and the tests shall be documented and repeatable, so that they can be re-run after any modification.
- The system documentation (SRS, design plan, class diagram, and database design) shall be kept up to date with any change made to the system.
- Updates and fixes shall be applicable without loss of existing data, including submitted versions, grades, and feedback.
- Semester changes (updating the lists of courses, students, and teachers) shall be done by the administrator through the system, with no programming effort.

3.3 System Functions

3.3.1 Sub-Functions for Authentication and Access Control

- ValidateLoginDetails(): Checks that the user's credentials are provided and correct before access is granted.
- SendVerificationCode(): Sends a verification code to the user's email as part of multi-factor authentication.
- VerifyAuthenticationCode(): Confirms the code entered by the user and grants access based on their role (Administrator, Teacher, or Student).
- CheckAccessRights(): Ensures each user can access only the functions and data permitted for their role, and only for their own courses.

3.3.2 Sub-Functions for User and Course Data Management

- AddTeacherRecord()/AddStudentRecord(): Allows the administrator to add student and teacher records with validated information.
- UpdateTeacherRecord()/UpdateStudentRecord: Allows the administrator to edit or remove outdated student and teacher information.
- AddCourseRecord(): Allows the administrator to add courses, including the assigned teacher.
- UpdateCourseRecord(): Allows the administrator to add courses, including the assigned teacher.
- EnrollStudentInCourse(): Links students to the courses they are registered in for the semester.

3.3.3 Sub-Functions for Course Material Management

- UploadCourseMaterial(): Allows teachers to upload lectures, notes, and slides for their courses, ensuring file type and size compliance.
- DisplayCourseMaterial(): Shows students the materials of their registered courses only.

- DownloadCourseMaterial(): Allows students to download the materials of their registered courses.

3.3.4 Sub-Functions for Assignment Management

- UploadAssignment(): Allows teachers to upload an assignment with its instructions, marks, and reference material.
- SetDeadline(): Records the submission deadline and allows it to be changed once.
- NotifyNewAssignment(): Sends students a notification when a new assignment is uploaded.

3.3.5 Sub-Functions for Assignment Submission and Version Control

- AttachFiles(): Manages the attachment of files (upload from device, photo, or link) and checks file type and the 2 MB size limit.
- OpenBlankDocument(): Opens a blank document in the system where the student can type their assignment and submit it as the assignment file.
- CheckFileFormat(): Detects non-PDF attachments and automatically notifies the student.
- ConvertToPDF(): Converts a non-PDF file to PDF when the student accepts the offer, and keeps the original file if conversion fails.
- SubmitAssignment(): Saves the submission, records the date and time, and displays a confirmation.
- VersionControlHandler(): Assigns a version number to each submission and archives previous versions unchanged.
- DisplayVersionHistory(): Shows students and teachers all versions of an assignment with their version number, date, time, and files.
- CheckDeadline(): Prevents submissions after the deadline has passed.

3.3.6 Sub-Functions for Grading and Feedback

- ViewSubmissions(): Allows teachers to view the submissions for their courses, including all versions.
- GradeSubmission(): Allows teachers to record the obtained marks for a selected version.
- SubmitFeedback(): Allows teachers to provide short text feedback on a version, where necessary.
- RetrieveGradeAndFeedback(): Displays the grade and feedback for each version to the student who submitted it.

4. Supplementary Material

4.1 Data Dictionary

Data Element Name	Definition	Data Type	Length	Example	Conditions
User Role	Role of the user in the system.	String	15	Student	Possible values: Administrator, Teacher, Student.
Password	Secret password used by the user to log in (stored in hashed form).	String	64	********	Required; must meet security standards; never stored in plain text.
Student Name	Full name of the student.	String	50	Mahnoor Shakeel	Required.
Student Email	Email address of the student.	String	50	student@example.com	Required; must be a valid email.
Student DOB	Date of birth of the student.	Date	10	15/03/2004	Required; must be in DD/MM/YYYY format.
Student Contact	Contact number of the student.	String	15	03001234567	Required; digits only.
Semester	Semester the student is currently studying in.	Integer	1	5	Required; value from 1 to 8.
Program	Degree program the student is enrolled in.	String	30	BS Computer Science	Required.
Department	Department the student belongs to.	String	50	Computer Sciences	Required.

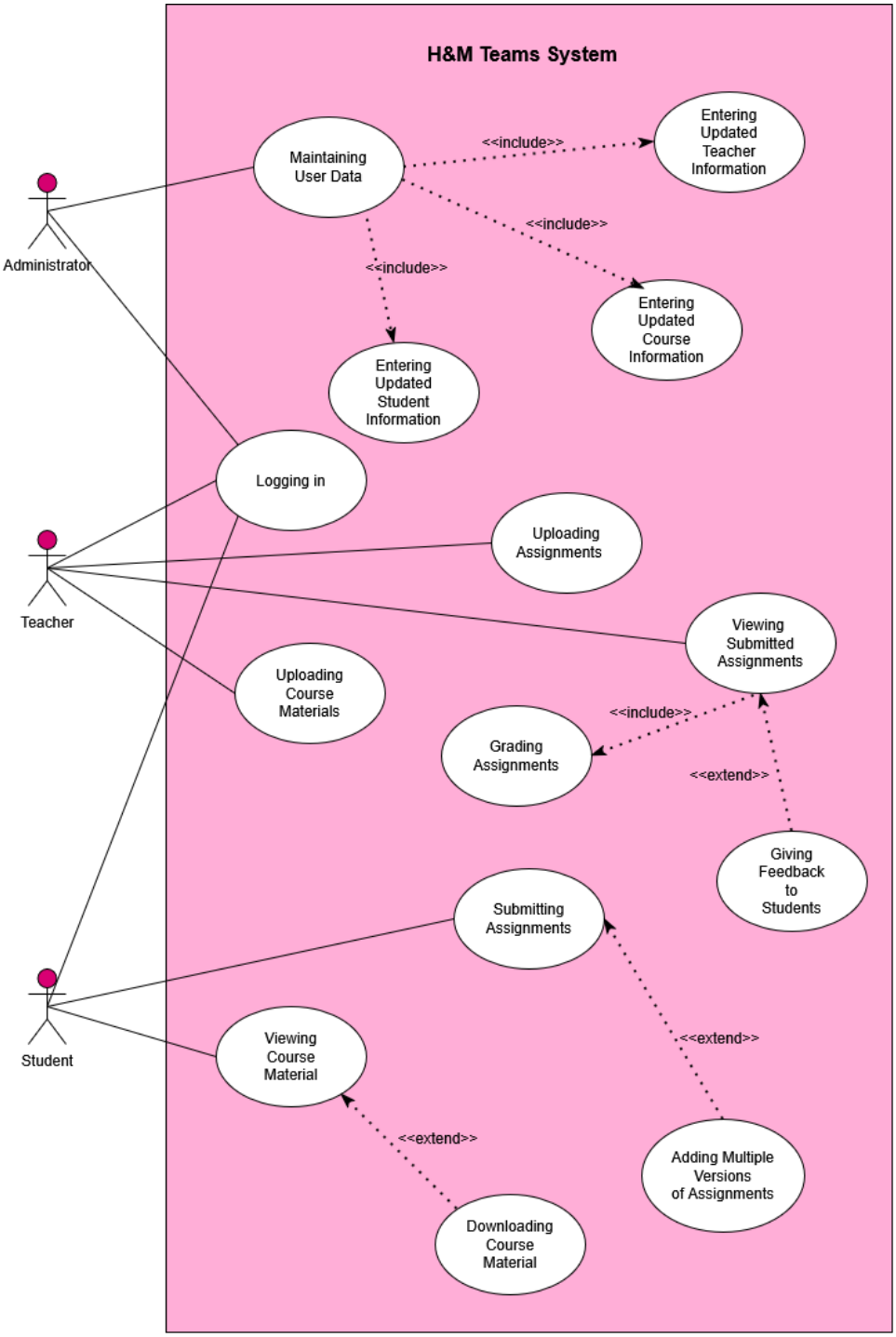
Data Element Name	Definition	Data Type	Length	Example	Conditions
Teacher Name	Full name of the teacher.	String	50	Dr. Onaiza Maqbool	Required; must exist in the system.
Teacher Email	Email address of the teacher.	String	50	teacher@example.com	Required; must be a valid email.
Administrator Name	Full name of the administrator.	String	50	Ali Raza	Required.
Course ID	Unique identifier for each course.	String	10	CS-322	Required; unique.
Course Name	Name of the course.	String	50	Software Construction	Required.
Course Description	Short description of the course.	String	300	Covers design and construction of software.	Optional.
Material ID	Unique identifier for each uploaded course material.	Integer	10	700123	Auto-generated unique ID.
Material Title	Title of the course material.	String	50	Lecture 3 Slides	Required.
Material File	File uploaded by the teacher as course material.	File	Up to 2 MB	Lecture3.pdf	Required; supported document formats only; maximum 2 MB.
Material Upload Date	Date when the material was uploaded.	Date	10	**25/09/2026**	Auto-generated; DD/MM/YYYY format.
Assignment ID	Unique identifier for each assignment.	Integer	10	310245	Auto-generated unique ID.

Data Element Name	Definition	Data Type	Length	Example	Conditions
Assignment Title	Title of the assignment.	String	50	Use Case Diagram	Required.
Assignment Instructions	Instructions given by the teacher for the assignment.	String	500	Draw the use case diagram for the system.	Optional.
Assignment Deadline	Date and time by which the assignment must be submitted.	Date and Time	16	09/10/2026 23:59	Required; DD/MM/YYYY HH:MM format; must be a future date when set.
Recipients	Students (registered in the course) to whom the assignment is sent.	String	200	CS-322 registered students	Required; only students registered in the course.
Marks	Maximum marks of the assignment.	Integer	3	20	Required; greater than 0.
Reference Material	Optional file attached by the teacher to help with the assignment.	File	Up to 2 MB	Reference.pdf	Optional; maximum 2 MB.
Submission ID	Unique identifier for each submission.	Integer	10	873922	Auto-generated unique ID.
Submission Version	Version number of the student's submission for an assignment.	Integer	2	2	Auto-generated; starts at 1 and increases with each new version; previous versions are retained.

Data Element Name	Definition	Data Type	Length	Example	Conditions
Submitted Files	File(s) submitted by the student (uploaded, photographed, or typed within the system).	File	Up to 2 MB per file	Assignment1.pdf	Required; supported formats only; maximum 2 MB per file.
Submission Date	Date when the student submitted the work.	Date	10	07/10/2026	Auto-generated; DD/MM/YYYY format.
Submission Time	Time when the student submitted the work.	Time	8	21:45:10	Auto-generated from server time; HH:MM:SS format.
Submission Status	Status of the submission.	String	15	Submitted	Possible values: Pending, Submitted, Failed.
Submission Confirmation	Message displayed after a successful submission.	String	30	✓ Submitted	Displayed only after the submission is saved.
Grade	Grade/marks given by the teacher to a submitted version.	Integer/String	3	17	Required; from 0 to the assignment's Total Marks.
Feedback	Short comments given by the teacher on the submitted version.	String	200	Good work, add more detail.	Optional.
Grading Date	Date when the grade was given.	Date	10	10/10/2026	Auto-generated; DD/MM/YYYY format.

Data Element Name	Definition	Data Type	Length	Example	Conditions
Notification ID	Unique identifier for each notification.	Integer	10	550321	Auto-generated unique ID.
Notification Message	Text shown to the student in the notification.	String	200	A new assignment has been uploaded.	Required.
Notification Date	Date when the notification was sent.	Date	10	07/10/2026	Auto-generated; DD/MM/YYYY format.

4.2 Use Case Diagram



4.3 Domain Model

